

Rylie Horning

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EDUCATION

- **Virginia Tech** Blacksburg, Virginia
B.S. Electrical Engineering, Minor in Computer Science
GPA 3.66/4.00 | Digital Systems (ECE 2544): 98%, Top 3 in class
August 2024 – May 2028
- **New River Community College** Dublin, Virginia
A.A.S. Computer Science, completed concurrently with high school
August 2021 – May 2024

SKILLS

- **Electrical Engineering:** Circuit Analysis, Circuit Design, Analog and Digital Electronics, LTspice Simulation, Schematic Design, PCB Assembly and Soldering, Sensors, Motor Controllers
- **Digital Systems and Programming:** Verilog/VHDL, Python (hardware testing), C/C++
- **Engineering Tools:** AutoCAD Electrical, SAP (ERP), Arduino, ESP32, Raspberry Pi, Trimble GPS
- **Professional:** Technical Documentation, Cross-Functional Engineering Support, Team Collaboration

EXPERIENCE

- **Moog Inc.** Christiansburg, Virginia
Supply Chain Intern
July 2025 - Present
 - Develop ISO 9001 compliant work instructions supporting supply movement processes.
 - Utilize SAP ERP systems to support engineering and production workflow documentation.
 - Attend daily stand-up meetings to give and receive progress updates.
- **Virginia Tech Electric Service** Blacksburg, Virginia
Engineering Intern
October 2024 - July 2025
 - Updated campus power distribution schematics and facility electrical maps using AutoCAD Electrical.
 - Collected electrical infrastructure data using Trimble GPS for asset mapping and documentation.
 - Assisted with installation and maintenance of campus electrical distribution equipment.
- **Motion Control Systems** Radford, Virginia
Electrical Engineering Intern
January 2024 - September 2024
 - Assembled and soldered motor control circuit boards for system testing and validation.
 - Implemented redline revisions to PCB schematics and layouts with engineers.
 - Developed Python scripts for hardware testing and simulation.
- **First Robotics Competition Team 401** Blacksburg, Virginia
Electrical Lead
August 2022 - May 2024
 - Led electrical design including power distribution, wiring harnesses, and sensor integration.
 - Created wiring schematics and electrical diagrams using Fritzing software.
 - Helped the team qualify for international competition, reaching quarterfinals.

PROJECTS

- **Graphic Equalizer for Home Audio System** — Analog Filters, Circuit Design
Designed a 3-band (bass, mid, treble) graphic equalizer covering the 100 Hz–8 kHz audio range using active band-pass filters and op-amp circuits. Modeled and verified filter responses in LTspice.
- **Automotive Interior LED System** — 12V DC Systems, Relays
Designed an automatic interior lighting system triggered by vehicle headlight signals using relay switching and fused power distribution.
- **Wi-Fi “On Air” LED Controller** — ESP32, Embedded Systems
Built a web-controlled ESP32 relay LED sign using embedded networking and relay control.